

Catalytic competence bounds the magnetic field effect of a thermally formed radical pair

Hikaru Wakaura^{*,†,‡} and Taiki Tanimae[†]

[†]*QIRI (Quantum Integrated Research Institute Inc.), Tokyo 107-0061, Japan*

[‡]*ORCID: 0000-0001-8381-8323*

E-mail: h.wakaura@qiri.co.jp

Abstract

A flavoenzyme could respond to the geomagnetic field through the radical-pair mechanism only if its catalytic cycle passes through a spin-correlated radical pair that lives long enough for a 50 μT field to act. For pairs created by light this is established; for pairs that would have to form thermally, as proposed for single-electron-transfer mechanisms of monoamine oxidase, it has been argued by analogy. Here we show that detailed balance turns catalytic competence into a lower bound on both decay rates of such a pair. If the pair lies Δ above the enzyme–substrate complex and carries the turnover k_{cat} , its singlet back-transfer rate and its spin-independent forward rate must both exceed $k_{\text{cat}} e^{\Delta/k_B T}$, for any spin Hamiltonian and without reference to electronic couplings or reorganization energies. Combining this bound with the largest field effect that any exchange coupling and field direction allow at given rates, computed for a flavin–aminium pair with density-functional hyperfine and dipolar couplings, gives the largest magnetic field effect on product formation compatible with turnover as a function of Δ . Over the dipolar couplings examined, the effect falls below 0.1 % for $\Delta \geq 0.47$ eV at $k_{\text{cat}} = 1 \text{ s}^{-1}$, and no pair above 0.86 eV can carry any turnover. Measured one-electron potentials place the flavin–amine pair of monoamine oxidase at least 1.61 eV above the ground state in water, so a magnetically sensitive radical pair of this kind would require the protein to narrow the amine–flavin potential gap by 0.75 V or more. In the same model a photochemically formed cryptochrome pair responds by 0.17–0.63 %.

Introduction

The radical-pair mechanism (RPM) is the only experimentally supported way in which a magnetic field as weak as the geomagnetic field ($\sim 50 \mu\text{T}$) can alter the yield of a chemical reaction.^{1,2} It underlies the leading model of avian magnetoreception, in which photoexcitation of the flavin in cryptochrome creates a spin-correlated flavin–tryptophan radical pair.^{3–5} Whether the same mechanism could act in flavoenzymes that work in the dark, such as the

monoamine oxidases that degrade neurotransmitters, has been raised repeatedly.^{6,7}

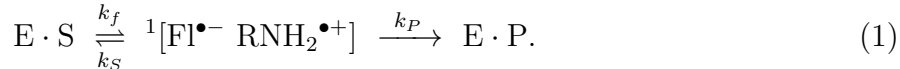
General limits on weak-field effects are known. Electron spin relaxation caps a primary RPM effect in biochemistry at 10^{-3} – 10^{-4} or less,⁸ and exchange and dipolar couplings between the radicals suppress singlet–triplet interconversion unless the radicals are far apart.⁹ Asymmetric reaction schemes, in which only the singlet pair recombines while both spin states react onward, relax the second limit and allow closely bound pairs to respond.¹⁰ What these results do not settle is whether a pair that an enzyme forms *thermally* can have the lifetimes that any of these mechanisms require. The exchange coupling and the recombination rate are usually treated as free parameters, and so is the rate of onward reaction.

Here we remove the rates as free parameters. A pair that is formed thermally from a closed-shell enzyme–substrate complex, and that must carry the catalytic flux, is subject to detailed balance, and we show that this fixes a lower bound on both of its decay rates in terms of the driving force Δ and the turnover number alone. We then compute the largest magnetic field effect that any exchange coupling allows at given rates for a flavin–aminium pair, combine the two into a bound on the field effect as a function of Δ , and locate the monoamine oxidase pair on it with measured redox potentials. Cryptochrome, treated in the same model, serves as a positive control.

Reaction scheme and spin model

Scheme and observable

We consider the single-electron-transfer (SET) scheme proposed for amine oxidation by flavoenzymes:¹¹



Electron transfer from the closed-shell complex forms the pair in its singlet state; only the singlet pair can return by back electron transfer (rate k_S), and both singlet and triplet pairs

react onward with a spin-independent rate k_P , for example by α -C–H deprotonation of the aminium radical. The observable that a change in catalytic flux would require is the yield of onward reaction per pair formed, $\Phi_P = k_P \int_0^\infty \text{Tr } \rho dt$, and we report the magnetic field effect as $\text{MFE}_P = [\Phi_P(B) - \Phi_P(0)]/\Phi_P(0)$ at $B = 50 \mu\text{T}$.

The spin density operator of a pair born in the singlet evolves as

$$\dot{\rho} = -2\pi i[H, \rho] - \frac{k_S}{2}\{P_S, \rho\} - k_P \rho, \quad \rho(0) = P_S/\text{Tr} P_S, \quad (2)$$

with P_S the singlet projector. Yields are obtained exactly, from the Sylvester equation for $\int \rho dt$ in the coherent case and in Liouville space when electron spin relaxation is included.

Spin Hamiltonian and conventions

In frequency units,

$$H = \omega_B \hat{n}_B \cdot (\mathbf{S}_1 + \mathbf{S}_2) + \sum_k a_k \mathbf{S}_{e(k)} \cdot \mathbf{I}_k + J \mathbf{S}_1 \cdot \mathbf{S}_2 + \mathbf{S}_1 \cdot \mathbf{T} \cdot \mathbf{S}_2, \quad (3)$$

with $\omega_B = g\mu_B B/h$ and isotropic hyperfine couplings a_k . With the exchange term written this way, $J = E_T - E_S$ is the full singlet–triplet gap; in the convention of Efimova and Hore⁹ $J = -2J_{\text{EH}}$. $\mathbf{T}$ is the traceless dipolar tensor; for an axial tensor $\mathbf{S}_1 \cdot \mathbf{T} \cdot \mathbf{S}_2 = 2D[(\mathbf{S}_1 \cdot \hat{n})(\mathbf{S}_2 \cdot \hat{n}) - \mathbf{S}_1 \cdot \mathbf{S}_2/3]$, so that with the field along $\hat{n}$ the $T_\pm$ and T_0 levels are split by D , the D of ref 9 ($D = -77.9$ MHz for point dipoles 1 nm apart). Both conventions are verified numerically in the code.

Magnetic parameters

For the flavin anion radical we use N5 (523 μT) and N10 (189 μT), each as a spin-1 nucleus.¹² For the aminium radical we use the two largest Fermi-contact couplings of the benzylaminium radical cation, 257 and 196 MHz (α -CH₂ protons; B3LYP/def2-TZVP on GFN2-xTB geome-

tries¹³). The same method gives the flavin N5 and N10 couplings within 16 % and 13 % of the values above. For the dipolar coupling of the contact pair we sum the point-dipole tensor over the Löwdin spin populations of the two radicals for methylamine stacked on the flavin si face at an N5–N distance of 3.5 Å; the resulting tensor has $D = -843$ MHz and $E = -26$ MHz. A pair separated further would have a smaller dipolar coupling, so we also compute every result with $\mathbb{T} = 0$, and with the contact tensor scaled by 0.1, 0.3 and 3 on the rates that set the thresholds below. Neither limit bounds the other (section), and the bound on the field effect is quoted for the largest value over all tensors computed.

Catalytic competence bounds both decay rates

Let Δ be the free energy of the radical pair relative to E·S, summed over its four spin states, so that the singlet pair lies $\Delta_S = \Delta + k_B T \ln 4$ above E·S (with $k_B T \ln 4 = 0.037$ eV at 310 K). Detailed balance for the singlet channel gives $k_f = k_S e^{-\Delta_S/k_B T}$, and the turnover carried by the pair is $k_f \Phi_P$ per E·S complex. Two inequalities bound Φ_P . Trivially $\Phi_P \leq 1$. Less obviously,

$$\Phi_P \leq 4k_P/k_S \tag{4}$$

for any spin Hamiltonian, with or without spin relaxation. To see this, note that the k_P term in eq 2 is proportional to the identity and commutes with the rest of the generator, so the steady-state pair population under a constant singlet source can only decrease when k_P is switched on. At $k_P = 0$ the steady state is the identity on the part of spin space reachable from the singlet, weighted by the singlet equilibrium constant; its trace is at most four times the singlet population. Eq 4 follows, and it is checked numerically for random spin systems with exchange, dipolar and hyperfine couplings and electron spin relaxation.

Requiring the turnover $k_f \Phi_P$ to be at least k_{cat} then gives

$$k_S \geq k_{\text{cat}} e^{\Delta/k_B T} \quad \text{and} \quad k_P \geq k_{\text{cat}} e^{\Delta/k_B T} \equiv k_{\min}(\Delta). \tag{5}$$

Without spin dynamics the turnover is exactly $e^{-\Delta_S/k_B T}/(1/k_S + 1/k_P)$, and eq 5 is its obvious consequence. Equation 5 contains no electronic coupling, no reorganization energy and no assumption about the chemistry of the onward step. Since no elementary rate exceeds the frequency of nuclear motion, $\nu_n \approx 10^{13}\text{--}10^{14} \text{ s}^{-1}$, it also bounds the driving force of any competent pair: $\Delta \leq k_B T \ln(\nu_n/k_{\text{cat}}) = 0.62\text{--}0.86 \text{ eV}$ for $k_{\text{cat}} = 10^3\text{--}1 \text{ s}^{-1}$.

The largest field effect at given rates

The exchange coupling of a contact pair is the sum of a reactive contribution from the same electron transfer that sets k_S ¹⁴ and a direct contribution that is not constrained by the rates. We therefore treat J as free and define

$$F(k_S, k_P) = \max_{J, \hat{n}_B} |\text{MFE}_P|. \quad (6)$$

J is scanned on a grid that is dense near zero and extends to $\max(4000 \text{ MHz}, 5(k_S + k_P)/2\pi)$, with repeated refinement around the three largest maxima, and $\hat{n}_B$ over 17 directions on a hemisphere. Cells whose maximum lies at the edge of the J range are recomputed with a range extended sixteenfold; none remains at the edge (0 of 195 cells with the contact tensor).

Figure 1 shows F over $k_S = 10^{-2}\text{--}10^5 \mu\text{s}^{-1}$ and $k_P = 10^{-2}\text{--}10^4 \mu\text{s}^{-1}$. The onward rate controls the response. With the contact dipolar tensor, $\max_{k_S} F$ is 1.4% at $k_P = 10^{-2} \mu\text{s}^{-1}$, 0.02% at $10^2 \mu\text{s}^{-1}$ and $1.1 \times 10^{-7} \%$ at $10^4 \mu\text{s}^{-1}$; without dipolar coupling the corresponding values are 14, 0.013 and $1.1 \times 10^{-7} \%$. The dipolar coupling mostly suppresses the response, but not everywhere: in 74 of the 195 cells, mostly at fast decay, the contact tensor gives a larger F than no dipolar coupling, by up to a factor of 8. A fast singlet back-transfer does not by itself remove the effect, because the triplet pair cannot return and is drained only by k_P ,¹⁰ but a fast spin-independent decay removes it for every k_S and every J .

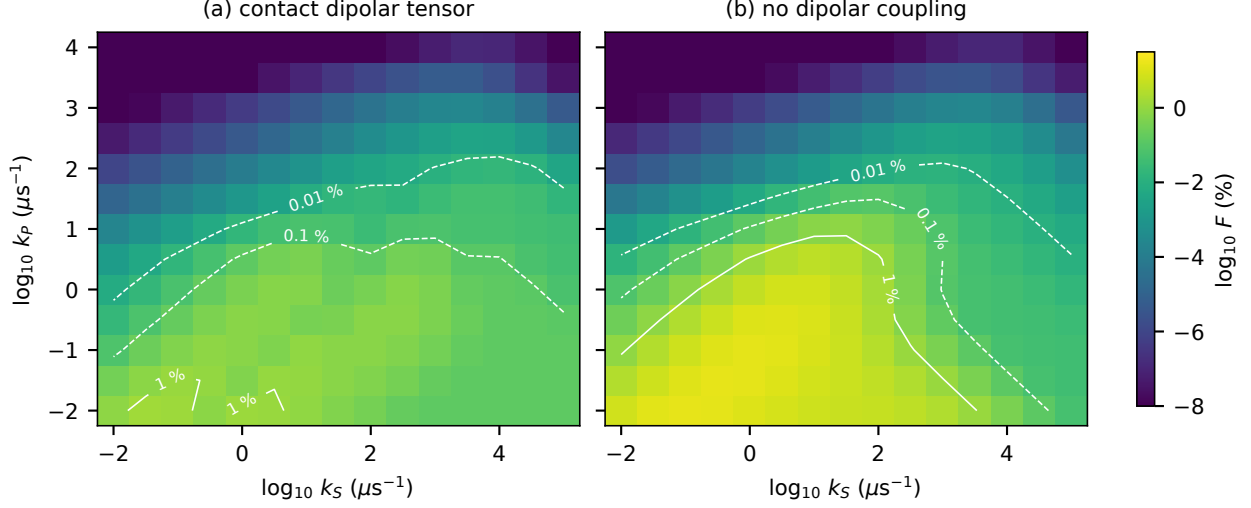


Figure 1: Largest magnetic field effect on the onward yield, maximized over the exchange coupling and the field direction, as a function of the singlet back-transfer rate k_S and the spin-independent onward rate k_P . (a) Contact dipolar tensor; (b) no dipolar coupling. Flavin N5 and N10 (spin 1) and two aminium protons; coherent spin dynamics; $B = 50 \mu T$.

A bound on the field effect of a competent pair

Combining eq 5 with F gives the largest field effect that a thermally formed pair can show while carrying the turnover,

$$G(\Delta) = \max\{ F(k_S, k_P) : k_S, k_P \geq k_{\min}(\Delta) \}. \quad (7)$$

On the rate grid, $k_{\min}$ is rounded down to the nearest grid rate in each direction, so that rates just above $k_{\min}$ are covered by the cell below them; above the largest k_P computed, G is bounded by its value at the grid edge because $\max_{k_S} F$ decreases monotonically with k_P on the grid, and above the largest k_S only the constraint on k_P is imposed. We evaluate G for the contact tensor, for no dipolar coupling, and for their cell-by-cell maximum together with the scaled tensors (the envelope). Figure 2 shows G for $k_{\text{cat}} = 1-10^3 \text{ s}^{-1}$. At $\Delta = 0$ and $k_{\text{cat}} = 1 \text{ s}^{-1}$ the bound is 1.4% with the contact tensor and 14% without dipolar coupling: a near thermoneutral pair is not excluded by competence. For the envelope the bound falls below 0.1% for $\Delta \geq 0.47 \text{ eV}$ and below 0.01% for $\Delta \geq 0.53 \text{ eV}$ at $k_{\text{cat}} = 1 \text{ s}^{-1}$ (0.44 and

0.53 eV with the contact tensor alone); at $k_{\text{cat}} = 10^3 \text{ s}^{-1}$ the 0.1 % threshold is 0.28 eV. Beyond $\Delta \approx 0.86 \text{ eV}$ no pair of this kind carries turnover at all.

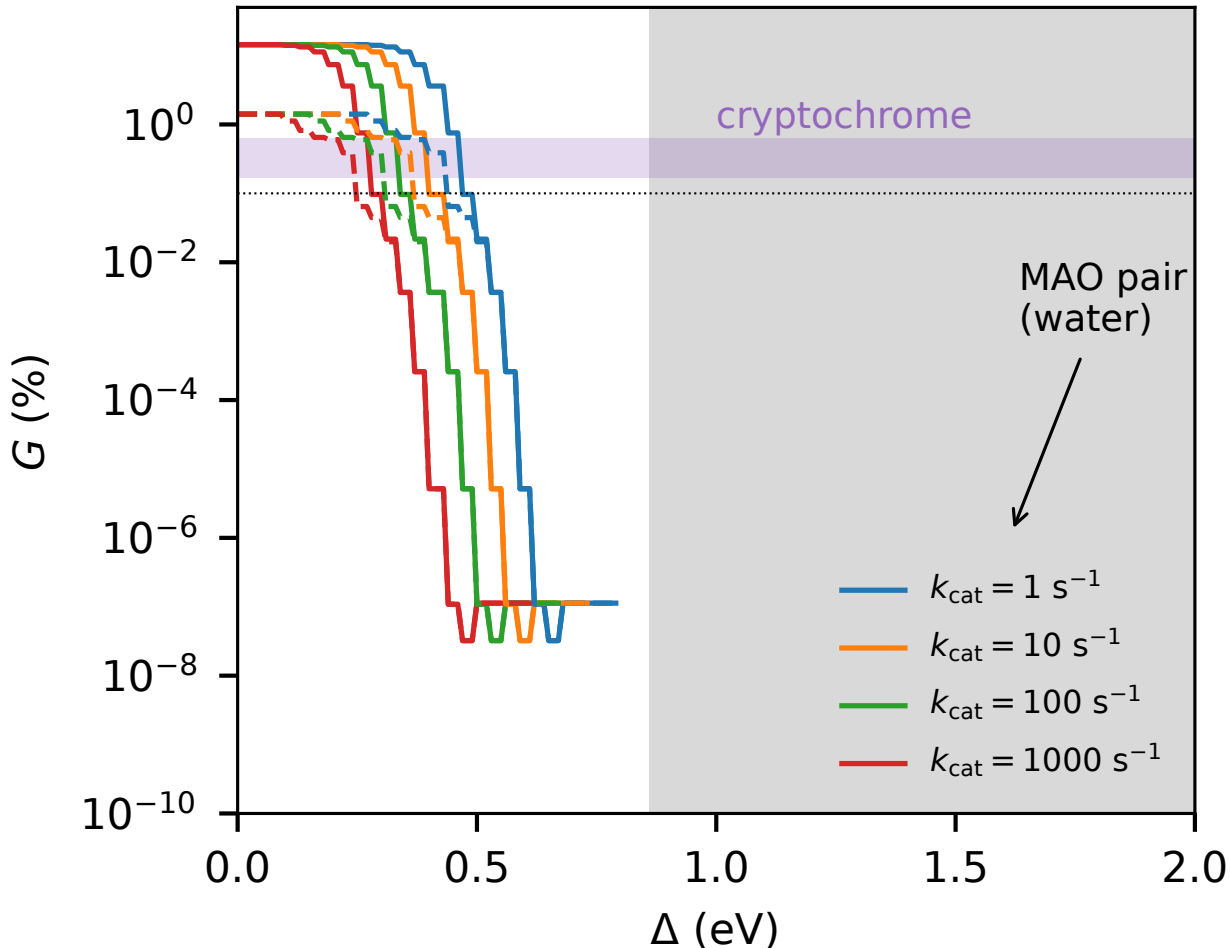


Figure 2: Largest magnetic field effect compatible with catalytic turnover, $G(\Delta)$, for $k_{\text{cat}} = 1, 10, 100$ and 10^3 s^{-1} . Solid: largest value over all dipolar tensors computed; dashed: contact dipolar tensor alone. Shaded: driving forces at which no thermal pair can carry the turnover. Arrow: lower bound on Δ for the monoamine oxidase flavin-amine pair from measured potentials in water. Band: cryptochrome in the same model.

Where monoamine oxidase lies

The driving force of the SET step is set by the one-electron potentials of the two couples. For dialkylamine radical cations in water $E^\circ(\text{R}_2\text{NH}^{\bullet+}/\text{R}_2\text{NH}) = 1.30 \pm 0.05 \text{ V vs NHE}$; ¹⁵ primary alkylamines, the substrates of monoamine oxidase, have higher gas-phase ionization

energies and are not easier to oxidize. The one-electron reduction of flavin mononucleotide to its semiquinone at pH 7 is at -0.313 V.¹⁶ In water the flavin–amine pair therefore lies at least $\Delta = 1.61$ eV above the closed-shell state (the contact Coulomb term in water is about 0.05 eV). This is well beyond the competence limit: the thermal formation rate cannot exceed $\nu_n e^{-\Delta/k_B T} \approx 6 \times 10^{-14} \text{ s}^{-1}$.

Our density-functional estimates are consistent with this but not more accurate. With the same protocol (B3LYP/def2-SVP, polarizable continuum) the two couples in water are in error by -0.57 V (dimethylamine) and -0.84 V (flavin), errors that largely cancel in Δ (net $+0.27$ eV); after removing these errors, methylamine and benzylamine give $\Delta \approx 2.1$ and 2.3 eV in water, and the raw values in a low-dielectric continuum ($\epsilon = 4$ – 10) with a contact Coulomb term are 2.47 – 2.99 eV. We use the measured potentials.

For the pair to be catalytically relevant, the protein would have to lower Δ by at least 0.75 eV relative to water; for G to exceed 0.1 % it would have to lower it further, to below 0.47 eV. Equivalently, with the amine couple unchanged, the one-electron flavin potential in the active site would have to lie above $+0.44$ V vs NHE. A measurement of the one-electron potential of the enzyme-bound flavin would test this directly. The kinetic evidence points the same way: no flavin radical is detected during the reductive half-reaction of MAO B,¹⁷ and the substituent effects and deuterium isotope effects of MAO A indicate that α -C–H bond cleavage is rate limiting and proceeds by proton abstraction.¹⁸

The electronic coupling between flavin and substrate does not enter the bound, but it is informative about the regime. Fragment-orbital couplings for methylamine at van der Waals contact (69 non-clashing geometries of 112; Supporting Information) are 53 meV (median) to 106 meV at 3.5 Å, large enough that back electron transfer at contact would be adiabatic, so that k_S is limited by nuclear motion rather than by the coupling.

Cryptochrome in the same model

The flavin-tryptophan pair of cryptochrome is formed by light, so eq 5 does not apply to it. Evaluated with the same scheme, the same observable and $k_S = k_P = 1 \mu s^{-1}$, with the pair 1.90 nm apart ($D = -11.4$ MHz), flavin N5 and N10 as above, the two largest β -proton couplings (60 and 56 MHz) and N1 of the tryptophan radical cation from the same density-functional method, and the exchange coupling scanned continuously over the literature range of its prefactor with both signs,⁹ its orientation-averaged response is 0.0015–0.63 %, and 0.17–0.63 % over the central part of that range. Electron spin relaxation with $T_2^e = 1 \mu s$ ¹⁹ reduces it to about 0.6 of its coherent value in a reduced spin system. The same Hamiltonian that gives these values for cryptochrome gives the bounds above for a thermal pair; the difference is how the pair is made.

Discussion

The bound applies to any radical pair that is formed thermally from a closed-shell state, returns to it only through the singlet, and must carry the catalytic flux. It does not depend on how the pair recombines or reacts onward, on the electronic coupling or on the reorganization energy. It does depend on the magnetic parameters through F . We computed F for four nuclei with coherent spin dynamics and for a discrete set of dipolar tensors. Additional nuclei and spin relaxation generally reduce low-field effects, but this is not a theorem, and the dipolar coupling can raise F at fast decay; F should be recomputed for other pairs. The turnover condition assumes that the pair is the only route to product; a pair that is a side branch of a faster polar pathway is not bounded by eq 5, but neither does its field effect change the catalytic flux.

Two routes around the bound should be named. Three-radical and scavenging mechanisms²⁰ restore sensitivity to strongly coupled pairs but still require the pair to survive until the third radical acts, which eq 5 limits in the same way. And an active site that lowers Δ to

near zero is not excluded: at small Δ competence allows lifetimes of microseconds and field effects of order a percent. The question for a particular enzyme is therefore where its radical pair lies in Δ , which is measurable, rather than whether a radical pair can in principle be magnetosensitive.

Conclusions

For a radical pair that an enzyme forms thermally and through which it turns over, detailed balance bounds the singlet back-transfer rate and the spin-independent onward rate from below by $k_{\text{cat}}e^{\Delta/k_BT}$. Combined with the largest field effect that any exchange coupling allows at given rates, this gives a bound on the magnetic field effect as a function of the driving force alone, which falls below 0.1 % for $\Delta \gtrsim 0.47$ eV. Measured redox potentials place the flavin-amine pair of monoamine oxidase at least 1.61 eV above the ground state in water, outside the range in which it could carry turnover at all, unless the protein shifts the potential gap by 0.75 V or more. Cryptochrome, whose pair is made by light, is not subject to the bound and responds by tenths of a percent in the same model.

Supporting Information Available

Coupling scan geometries and fits; hyperfine couplings of all nuclei; the dipolar tensor from spin populations; driving-force calculations and the comparison with measured potentials; the full worst-case maps; the cryptochrome scan; software and environment.

Data availability

All code and numerical results are available in the open-source `qbsscreen` package at https://github.com/deeptell-inc/brain_protein_screening; every number in the text is generated from the stored results by `qbsscreen/make_numbers_v5.py`.

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TOC Graphic

